BABU BANARASI DAS UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Student Management System Using MongoDB

Submitted to:
Mr. Harendra Singh
Department: BCA (DS & AI)

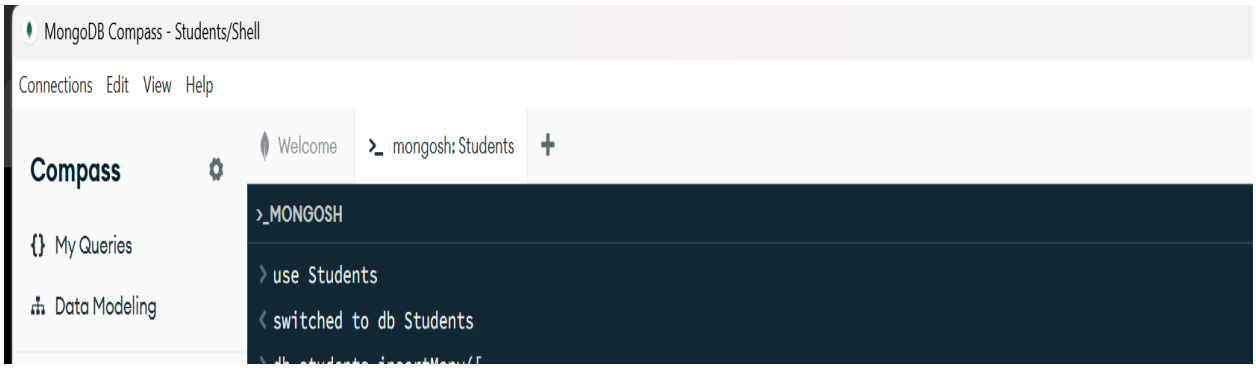
Submitted by:
Anchal Dubey
Roll no: (09)
1250258083

1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.

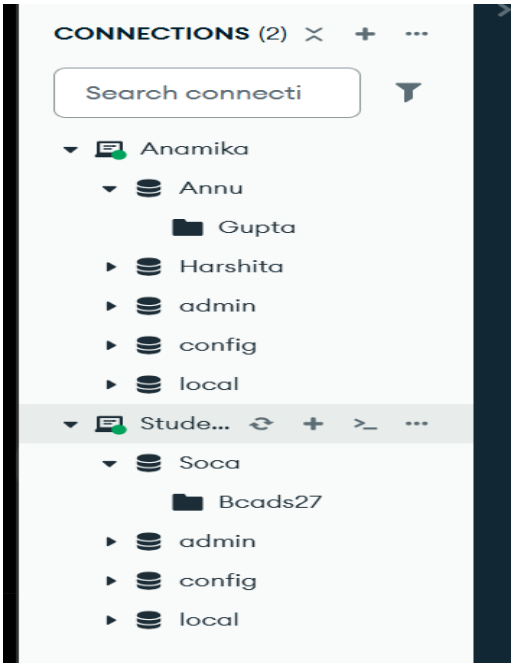


2. CREATE COLLECTION

MongoDB Query:

db. create Collection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

db. students. Insert Many([ ])

{roll:1, name:"Anamika", age:19,marks:88,city:"Lucknow"},

{roll:2, name:"Uday", age:18,marks:76,city:"Jaunpur"},

{roll:3, name:"Vaishnavi", age:19,marks:92,city:"Kanpur"},

{roll:4, name:"Annu", age:18,marks:69,city:"Ayodhya"},

{roll:5, name:"Vansh", age:19,marks:81,city:"Varanasi"},

{roll:6, name:"Amrita", age:20,marks:85,city:"Lucknow"},

{roll:7, name:"Naveen", age:19,marks:73,city:"Prayagraj"},

{roll:8, name:"Pratik", age:19,marks:90,city:"Lucknow"},

{roll:9, name:"Sangita", age:20,marks:78,city:"Gorakhpur"},

{roll:10, name:"Vivek", age:18,marks:65,city:"Kanpur"},

{roll:11, name:"Tarkeshwar", age:20,marks:71,city:"Jaunpur"},

{roll:12, name:"Shreya", age:19,marks:95,city:"Lucknow"},

{roll:13, name:"Akshita", age:18,marks:84,city:"Ayodhya"},

{roll:14, name:"Shruti", age:19,marks:87,city:"Prayagraj"},

{roll:15, name:"Kartik", age:20,marks:74,city:"Varanasi"},

{roll:16, name:"Sejal", age:18,marks:89,city:"Lucknow"},

{roll:17, name:"Astha", age:19,marks:82,city:"Kanpur"},

{roll:18, name:"Jai", age:18,marks:58,city:"Gonda"},

{roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},

{roll:20, name:"Omkar", age:19,marks:63,city:"Bareilly"},

{roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"},

{roll:22, name:"Saumya", age:19,marks:86,city:"Jaunpur"},

{roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},

{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},

{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"},

{roll:26, name: "Anushka", age:20,marks:94,city:"Lucknow"},

{roll:27, name: "Ananya", age:19,marks:88,city:"Varanasi"},

{roll:28, name: "Bhaskar", age:18,marks:61,city:"Gorakhpur"},

{roll:29, name: "Rohan", age:19,marks:75,city:"Kanpur"},

{roll:30, name: "Nishant", age:20,marks:79,city:"Lucknow"},

{roll:31, name: "Vineet", age:18,marks:68,city:"Jaunpur"},

{roll:32, name: "Anika", age:19,marks:90,city:"Lucknow"},

{roll:33, name: "Jai", age:20,marks:55,city:"Ayodhya"},

{roll:34, name: "Lakshya", age:18,marks:83,city:"Kanpur"},

{roll:35, name: "Krishna", age:19,marks:96,city:"Prayagraj"},

{roll:36, name: "Rumana", age:20,marks:70,city:"Lucknow"},

{roll:37, name: "Priyanshu", age:18,marks:81,city:"Varanasi"},

{roll:38, name: "Aditya", age:19,marks:66,city:"Gorakhpur"},

{roll:39, name: "Dhruv", age:20,marks:93,city:"Lucknow"},

{roll:40, name: "Nivedita", age:19,marks:85,city:"Kanpur"},

{roll:41, name: "Meenu", age:18,marks:74,city:"Jaunpur"},

{roll:42, name: "Akanksha", age:20,marks:89,city:"Lucknow"},

{roll:43, name: "Sarika",age:19,marks:62,city:"Ayodhya"},

{roll:44, name: "Anchal",age:18,marks:78,city:"Prayagraj"},

{roll:45, name: "Umra", age:20,marks:84,city:"Lucknow"},

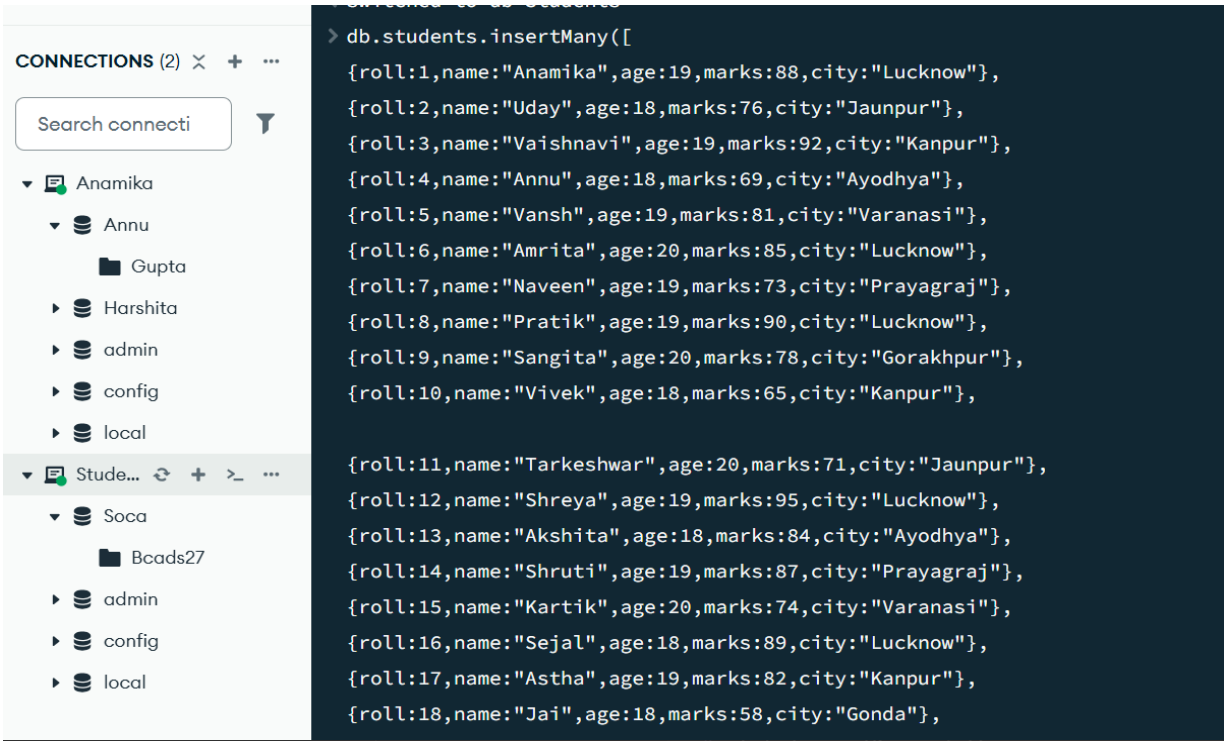
{roll:46, name: "Nashra", age:19,marks:73,city:"Kanpur"},

{roll:47, name: "Aman", age:18,marks:59,city:"Gonda"},

{roll:48, name: "Priya", age:20,marks:87,city:"Lucknow"},

{roll:49, name: "Rahul", age:19,marks:64,city:"Bareilly"},

{roll:50, name: "Neha", age:18,marks:91,city:"Varanasi"}



Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

db. students. Find ()

Explanation: Retrieves documents from the students collection.

```
> db.students.find()
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221d'),
    roll: 2,
    name: 'Uday',
    age: 18,
    marks: 76,
    city: 'Jaunpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
```

5. EQUAL TO — $eq

MongoDB Query:

db. students. Find ({marks: {$eq: 88}})

Explanation: Finds students whose marks are exactly 88.

```
> db.students.find({marks: {$eq: 88}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2236'),
    roll: 27,
    name: 'Ananya',
    age: 19,
    marks: 88,
    city: 'Varanasi'
  }
```

6. GREATER THAN — $gt

MongoDB Query:

db.students.find({marks: {$gt: 85}})

Explanation: Finds students whose marks are greater than 85.

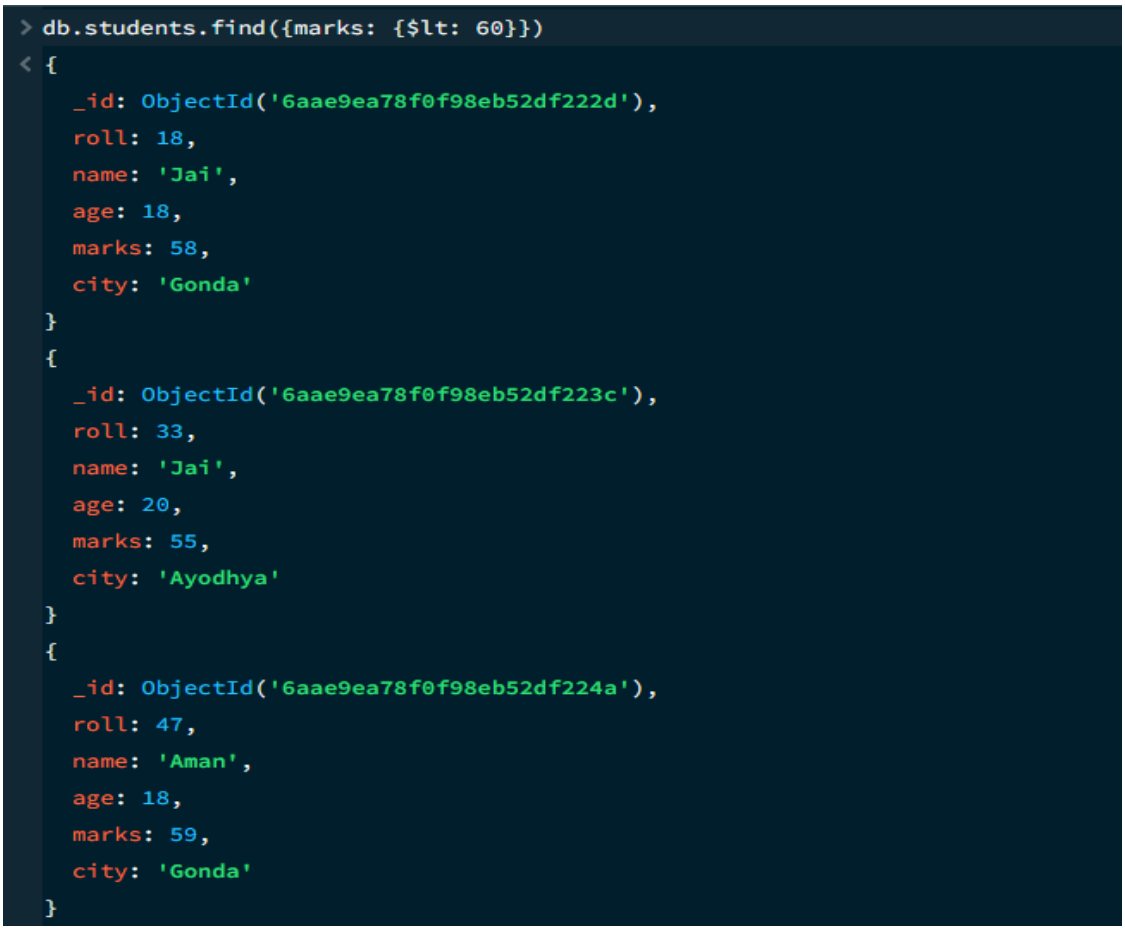
```
> db.students.find({marks: {$gt: 85}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
```

7. LESS THAN — $lt

MongoDB Query:

db. students. find ({marks: {$lt: 60}})

Explanation: Finds students whose marks are less than 60.



```
> db.students.find({marks: {$lt: 60}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df222d'),
    roll: 18,
    name: 'Jai',
    age: 18,
    marks: 58,
    city: 'Gonda'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df223c'),
    roll: 33,
    name: 'Jai',
    age: 20,
    marks: 55,
    city: 'Ayodhya'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df224a'),
    roll: 47,
    name: 'Aman',
    age: 18,
    marks: 59,
    city: 'Gonda'
  }
```

8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

db. students. find({marks: {$gte: 90}})

Explanation: Finds students whose marks are 90 or above.

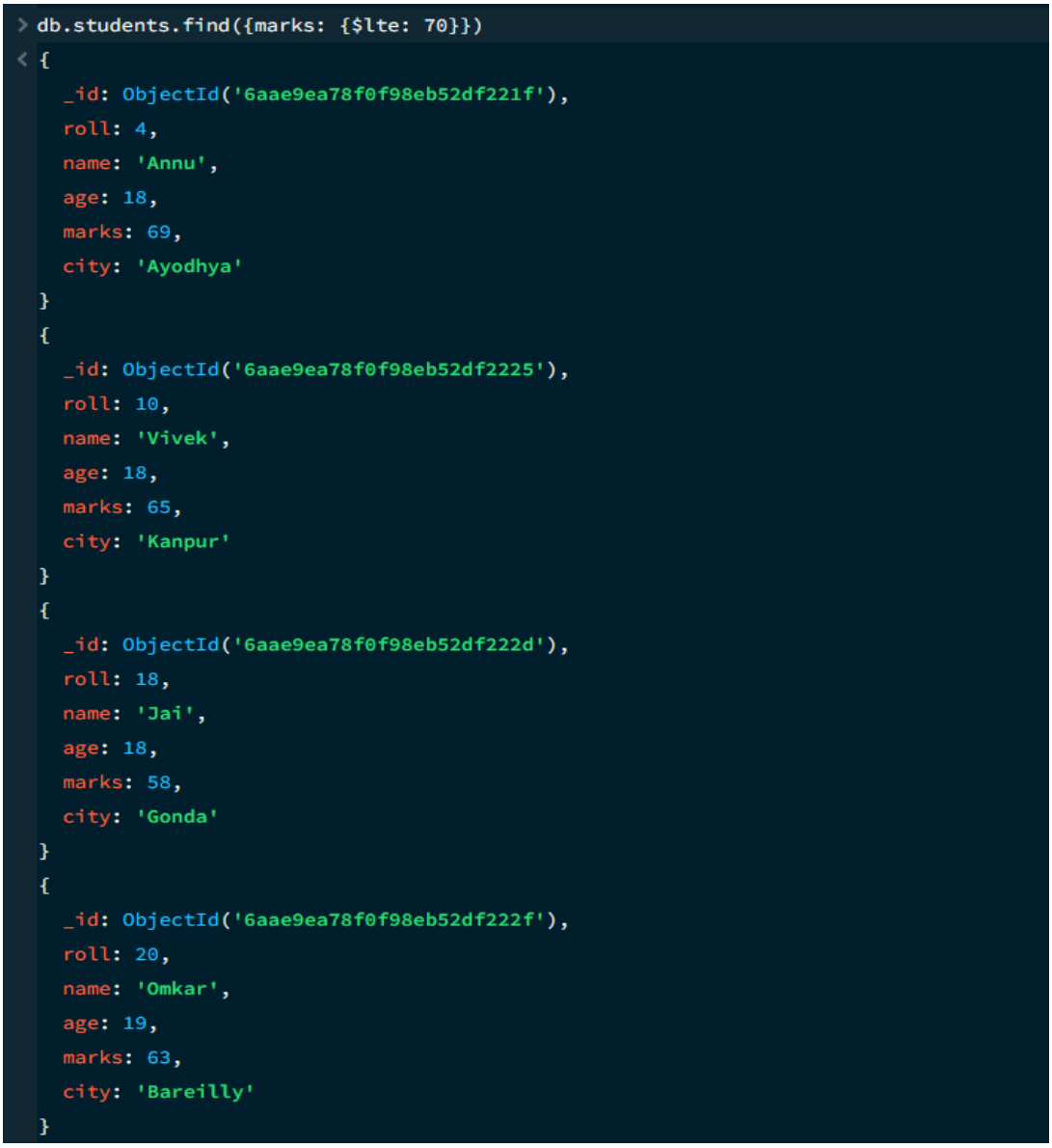
```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222e'),
    roll: 19,
    name: 'Harshita',
    age: 20,
    marks: 91,
    city: 'Lucknow'
  }
```

9. LESS THAN OR EQUAL TO — $lte

MongoDB Query:

db. students. find({marks: {$lte: 70}})

Explanation: Finds students whose marks are 70 or below.



```
> db.students.find({marks: {$lte: 70}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221f'),
    roll: 4,
    name: 'Annu',
    age: 18,
    marks: 69,
    city: 'Ayodhya'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2225'),
    roll: 10,
    name: 'Vivek',
    age: 18,
    marks: 65,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222d'),
    roll: 18,
    name: 'Jai',
    age: 18,
    marks: 58,
    city: 'Gonda'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222f'),
    roll: 20,
    name: 'Omkar',
    age: 19,
    marks: 63,
    city: 'Bareilly'
  }
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(
  {roll:5},
  {$set:{marks:95}}
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.

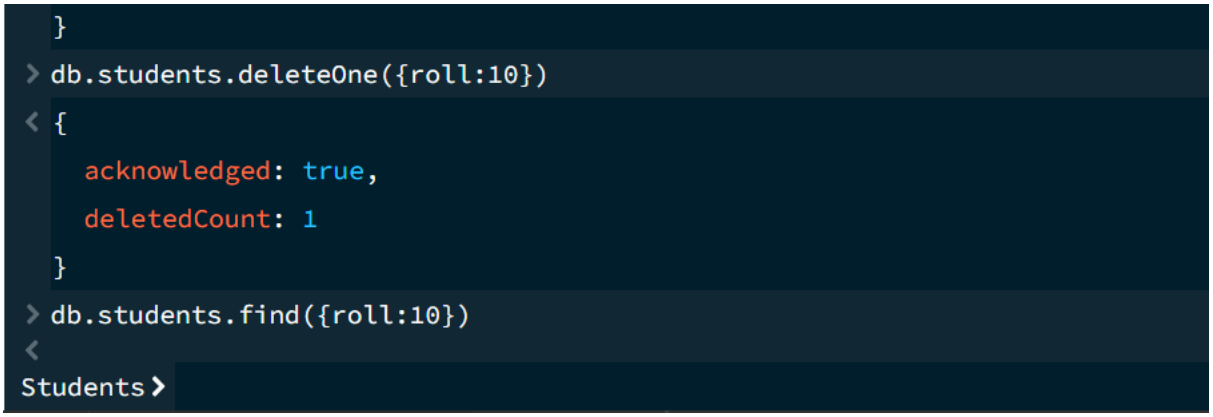


11. DELETE ONE OPERATION

MongoDB Query:

db.students.deleteOne({roll:10})

Explanation: Deletes one document matching roll number 10.

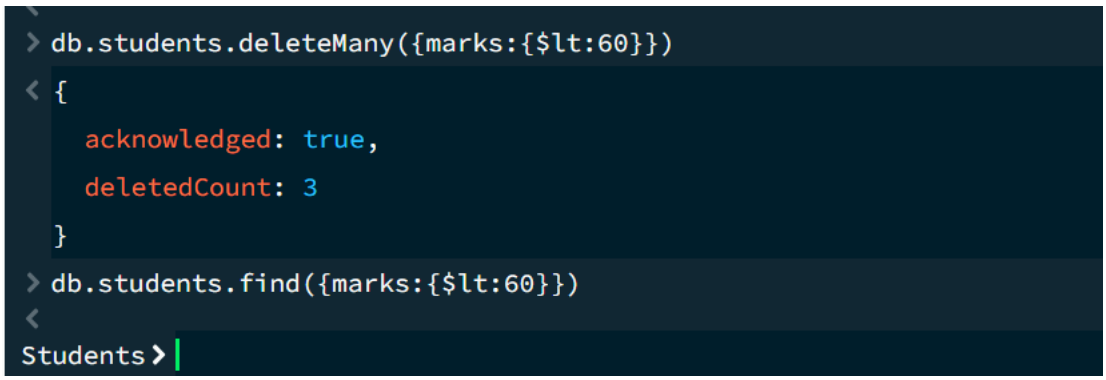


12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({
  marks: {$lt:60}
})
```

Explanation: Deletes all students having marks below 60.



13. AND OPERATION — $and

MongoDB Query:

```
db. students. find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

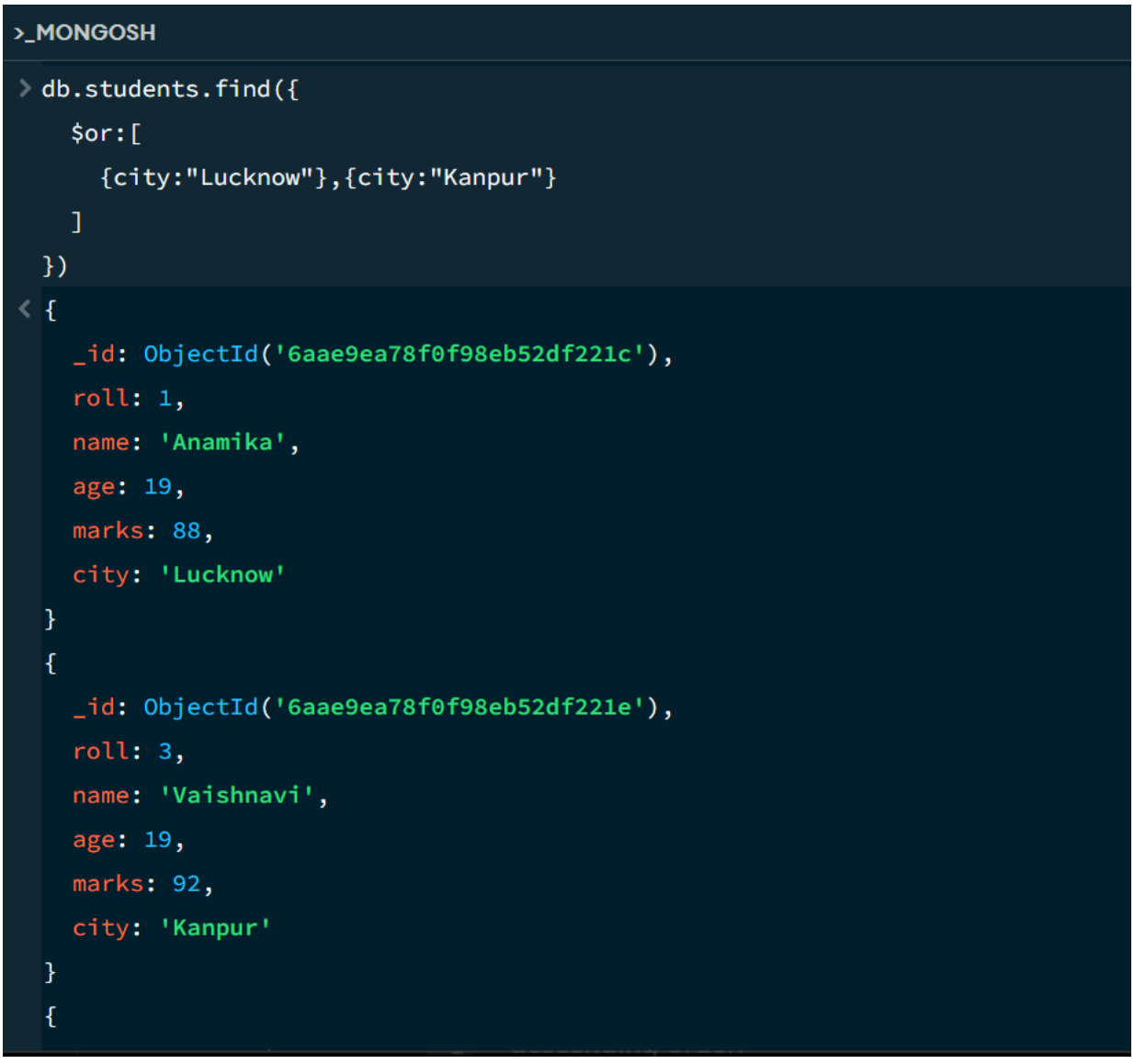
```
>_MONGOSH
> db.students.find({
    $and:[
      {marks:{$gt:80}},{age:19}
    ]
  })
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
```

14. OR OPERATION — $or

MongoDB Query:

```
db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.



15. SORT OPERATION

A.MongoDB Query:

db. students. find (). sort({marks : 1})

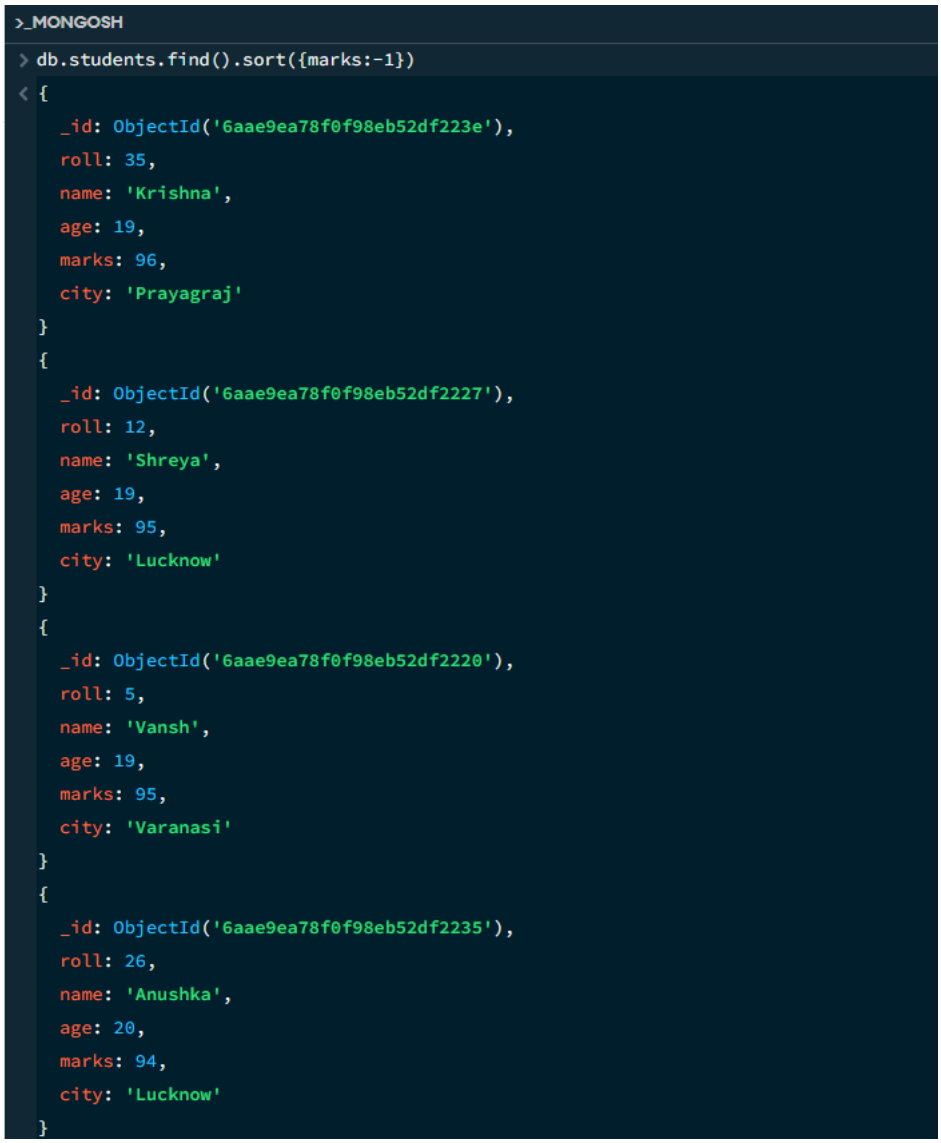
Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

```
>_MONGOSH
> db.students.find().sort({marks:1})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df2237'),
    roll: 28,
    name: 'Bhaskar',
    age: 18,
    marks: 61,
    city: 'Gorakhpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2246'),
    roll: 43,
    name: 'Sarika',
    age: 19,
    marks: 62,
    city: 'Ayodhya'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222f'),
    roll: 20,
    name: 'Omkar',
    age: 19,
    marks: 63,
    city: 'Bareilly'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df224c'),
    roll: 49,
    name: 'Rahul',
    age: 19,
    marks: 64,
    city: 'Bareilly'
  }
```

B. MongoDB Query:

db. students. find (). Sort ({marks: -1})

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.



```
>_MONGOSH
> db.students.find().sort({marks:-1})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df223e'),
    roll: 35,
    name: 'Krishna',
    age: 19,
    marks: 96,
    city: 'Prayagraj'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2220'),
    roll: 5,
    name: 'Vansh',
    age: 19,
    marks: 95,
    city: 'Varanasi'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2235'),
    roll: 26,
    name: 'Anushka',
    age: 20,
    marks: 94,
    city: 'Lucknow'
  }
```

16. LIMIT OPERATION

MongoDB Query:

db. students. find (). limit (5)

Explanation: Displays only the first five documents.

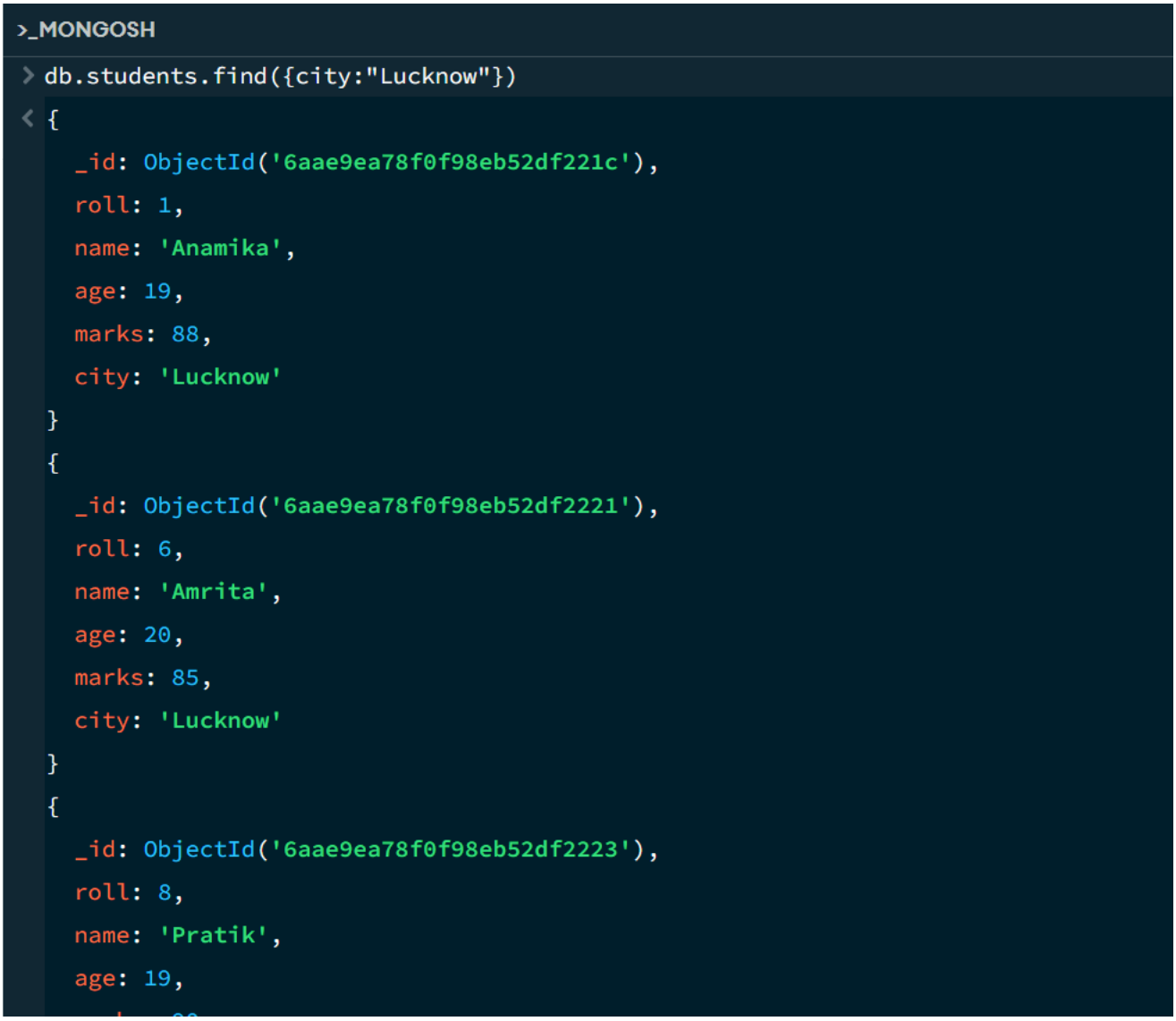
```
>_MONGOSH
> db.students.find().limit(5)
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221d'),
    roll: 2,
    name: 'Uday',
    age: 18,
    marks: 76,
    city: 'Jaunpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({city:"Lucknow"})

Explanation: Finds all students whose city is Lucknow.

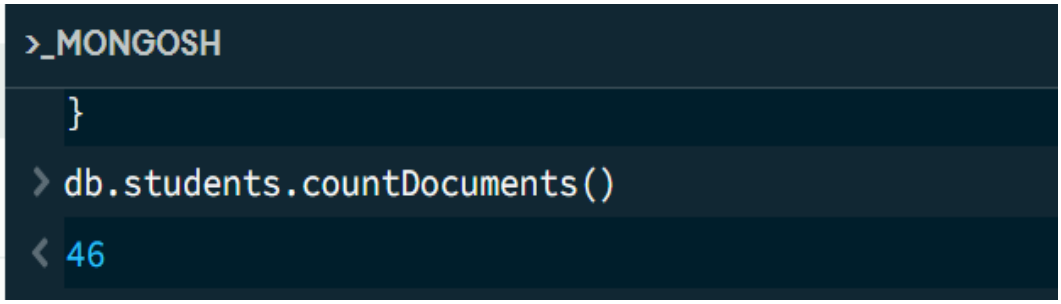


18. COUNT DOCUMENTS

MongoDB Query:

db. students. count Documents ()

Explanation: Counts documents currently present in the collection (performing after the delete function).



MONGODB OPERATORS USED:

Operator	Meaning	Example
$eq	Equal to	marks: {$eq:88}
$gt	Greater than	marks: {$gt:85}
$lt	Less than	marks: {$lt:60}
$gte	Greater than or equal to	marks: {$gte:90}
$lte	Less than or equal to	marks: {$lte:70}
$and	All conditions true	$and:[...]
$or	At least one condition true	$or:[...]